

Program : Diploma in Automation and Robotics	
Course Code: 6337	Course Title: Virtual Instrumentation Lab using LabView
Semester: 6	Credits: 2.5
Course Category: Program core	
Periods per week: 4 (L:0 T:0 P:4)	Periods per semester: 60

Course Objectives:

- Understand the components of a Virtual Instrument
- Introduce LabVIEW and common LabVIEW functions

Course Outcomes:

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Lab view Basics	15	Understanding
CO2	Explore Debugging features	15	Applying
CO3	To Identify tool set features	15	Applying
CO4	Applications	12	Applying
	Series Test	3	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3	2					
CO3	3						
CO4	3						

3-Strongly mapped

2-Moderately mapped

1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Labview Basics		
M1.01	To Familiarize with Virtual Instrumentation software Environment – LabView	4	Understanding.
M1.02	To practice Lab View Editing Techniques	4	Understanding.
M1.03	To implement a VI to convert Degree Celsius to Fahrenheit	2	Applying
M1.04	To implement Logic Gates using Labview	2	Applying
M1.05	To Construct basic ON-OFF control	3	Applying
CO2	Explore Debugging features		
M2.01	To Practice debugging features of Lab View	6	Applying
M2.02	To Implement VI with two blinking indicator lights with a delay of two seconds	3	Applying
M2.03	To create a case structure using switch	1	Applying
M2.04	To Create a simple calculator using Labview	3	Applying
M2.05	To create VI subsets and to convert VI to VI subset	2	Applying
	Series Test – I	1.5	
CO3	VI tool set.		
M3.01	To practice working with VI Tool Set	6	Applying
M3.02	To create a program to plot data stored in an array.	6	Applying
M3.03	To use a While Loop and a chart for acquiring and displaying data in real time	3	Applying
CO4	Applications		
M4.01	To measure current, voltage capacitance, resistance and inductance using ELVIS.	3	Applying

M4.02	To measure voltage drop across a resistor in a potential divider circuit	3	Applying
M4.03	To measure Temperature using Temperature sensor and DAQ module in Labview.	3	Applying
M4.04	Digital control of Stepper Motor	3	Applying
	Series Test – II	1.5	

Text / Reference

T/R	Book Title/Author
T1	LabVIEW programming,data acquisition and analysis , Beyon Jeffery Y, National instrument Virtual instrument series.
R1	PC Interfacing for Data Acquisition and Process Control, Instrument society of America, 1994
R2	Learning with Lab-view, Bishop Robert H, Prentice Hall
R3	Lab View for data acquisition, Brace Muhura, National instrument Virtual instrument series.
R4	Jovitha Jerome, Virtual Instrumentation using lab view, PHI learning pvt.ltd